

OPTICS

3D-printed multilayer structures for high-numerical aperture achromatic metalenses

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Flat optics consisting of nanostructures of high-refractive index materials produce lenses with thin form factors that tend to operate only at specific wavelengths. Recent attempts to achieve achromatic lenses uncover a trade-off between the numerical aperture (NA) and bandwidth, which limits performance. Here, we propose a new approach to design high-NA, broadband, and polarization-insensitive multilayer achromatic metalenses (MAMs). We combine topology optimization and full-wave simulations to inversely design MAMs and fabricate the structures in low-refractive index materials by two-photon polymerization lithography. MAMs measuring 20 μm in diameter operating in the visible range of 400 to 800 nm with 0.5 and 0.7 NA were achieved with efficiencies of up to 42%. We demonstrate broadband imaging performance of the fabricated MAM under white light and RGB narrowband illuminations. These results highlight the potential of the 3D-printed multilayer structures for realizing broadband and multifunctional meta-devices with inverse design.

INTRODUCTION

Recent progress in metalenses at micro- and macro-length scales have shown unprecedented potential to achieve remarkable imaging performance, finding applications in areas such as light-field imaging, bioanalysis, medicine, semiconductor, and quantum technologies. In particular, achromatic lenses exhibit broadband responses that enable capture of color information, which expands design possibilities and application scenarios for photonic devices (1–12). The desirable characteristics, e.g., ultra-compact, ultra-thin, and light-weight characteristics, make metalenses compelling in imaging systems (13–19). However, most metalenses are patterned in high-refractive index (HRI) materials that provide good optical control due to strong light confinement but are highly dispersive, making broadband implementations challenging.

The Abbe number is a figure of merit in lens design that represents how dispersion free a transparent material is, i.e., higher values indicate lower dispersion. Commonly used HRI materials in metalenses such as Si, TiO₂, and GaN have Abbe numbers of 5.5, 9.86, and 18.7, respectively. In contrast, the Abbe number of a typical resin IP-Dip with a refractive index of ~ 1.56 is ~ 35.2 (details in fig. S1). As HRI materials are highly dispersive and the subwavelength nanoantennae also induce strong scattering in such meta-devices, the concomitant realization of metalenses with (i) high numerical aperture (NA), (ii) broadband (achromatic), and (iii) high efficiency in the visible range is hindered. Some of the recent works on achromatic metalenses tackle these three points in the near- or

mid-infrared (IR) range, where the material dispersion is substantially lower and material absorption is negligible (20–22).

Single lenses based on traditional refractive optics and diffractive optics exhibit chromatic aberrations, as shown in Fig. 1B. Single-layer multilevel diffractive lenses (MDLs) (23) and metalenses minimize chromatic aberration at the expense of low NA, due to the intrinsic trade-off between bandwidth and NA for a given lens radius (14),

$$R_{\text{max}} \leq \frac{C}{\Delta\omega \left(\frac{1}{\text{NA}} - \sqrt{\frac{1}{\text{NA}^2} - 1} \right)} \quad (1)$$

where R_{max} is the maximum achievable radius of the lens; C is a constant related to the material, central frequency, and structure; and $\Delta\omega$ is the bandwidth. The premise of Eq. 1 is that the phase distribution of the generated wavefront meets the requirement of an aberration free metalens (24). Therefore, this formula is the relationship that needs to be satisfied to realize a high-efficiency focusing lens. It can be further derived that for a given lens radius, the maximum bandwidth is inversely proportional to NA, i.e., $\Delta\omega \propto \frac{C_1}{\text{NA}}$. C_1 is also a constant related to the material, central frequency, and structure.

Multilayer metalenses are not constrained by Eq. 1, which holds only for the case of single-layer metalenses. In conventional refractive lenses, the lens doublet is often used to reduce chromatic aberrations, which uses two different materials and geometric designs to compensate the dispersive response. Similarly, multilayer metalenses have been explored to introduce additional degrees of freedom (DOFs) (25–28) because of the constraints in simultaneously improving NA and bandwidth in single-layer metalenses. With the additional DOF available for the design of multilayer structures, enhanced abilities of light manipulation by wavefront manipulation or even dynamic changes by multilayer metadevices become possible (29). By using synergistic effects between the individual layers, the multilayer structure can be designed to break the design trade-off between high NA and bandwidth (Fig. 1C).

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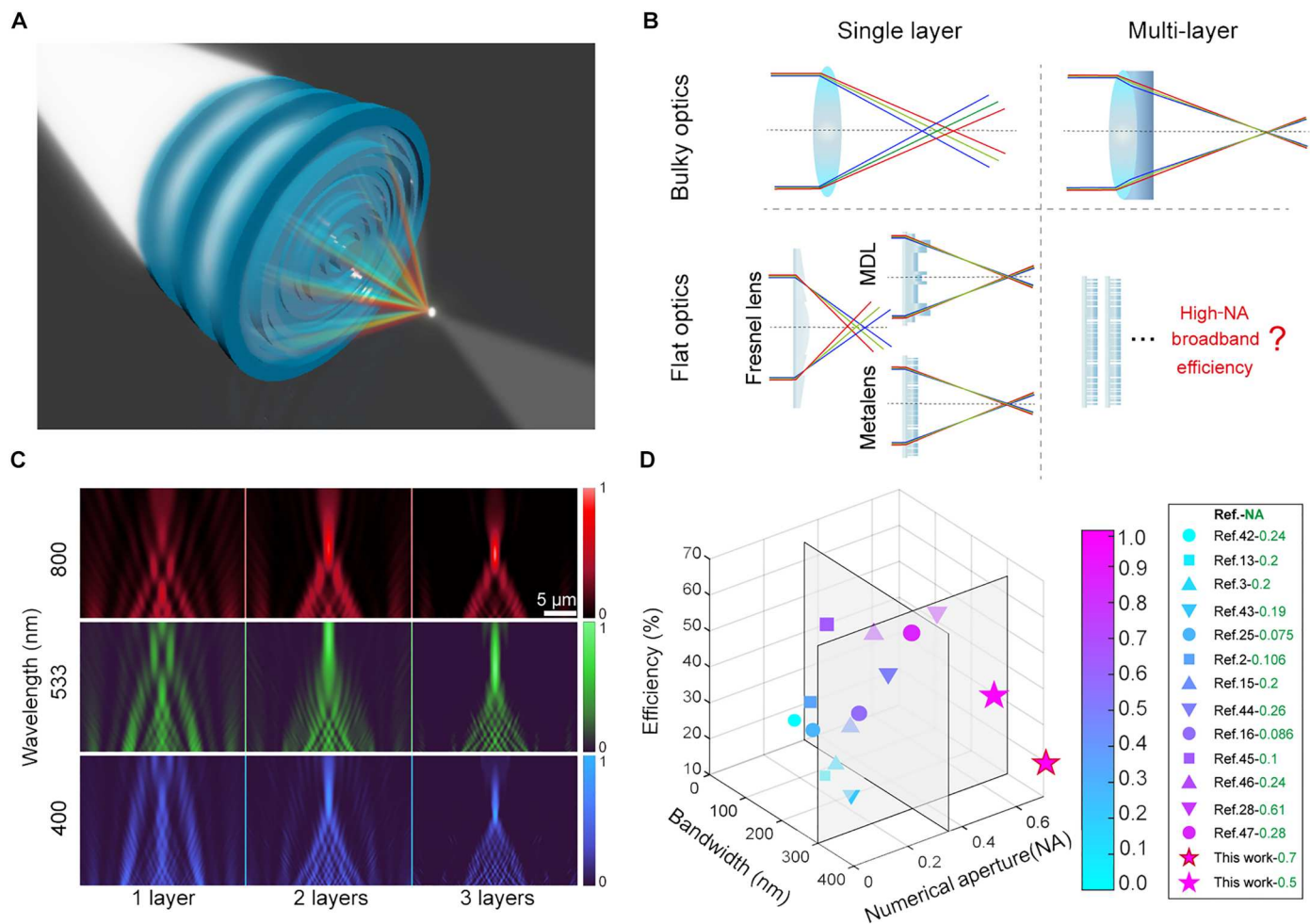


Fig. 1. Achromatic performance of different single lenses and multilayer lenses. (A) Schematic of the designer 3D-printed multilayer achromatic metalens (MAM). (B) Schematic of traditional and flat optics lenses [including Fresnel lens, multilevel diffractive lens (MDL), and metalens] with single layers and multilayers. (C) Evolution of focal spots at wavelengths of 400, 533, and 800 nm when additional layers are added (results are from an optimized three-layer design of 0.5-NA MAM). (D) The efficiencies, numerical apertures, and bandwidths (works in the visible band) of various achromatic metalenses (details in tables S1 and S2) (2, 3, 13, 15, 16, 25, 28, 47, 51–55). The color bar and marker size represent the figure of merit defined as the square root of the sum of squares of efficiency, NA, and bandwidth. The gray planes indicate previous limits at bandwidth = 300 nm and NA = 0.35. The NA values of each metalens are illustrated in the legend.

The fabrication challenges associated with producing multilayer achromatic metalenses (MAMs) are lifted using nanoscale three-dimensional (3D) printing. Due to the high-resolution patterning requirements, electron beam lithography (EBL) has been widely used to fabricate nano-scale meta-devices including metalenses. However, patterning multilayer and variable height structures using EBL is inconceivable, requiring alignment and multistep fabrication processes. Conversely, nanoscale 3D printing enables the patterning of a multilayer metalens in one lithographic step, which facilitates rapid prototyping of complicated structures. With two-photon polymerization lithography (TPL), various 3D designs have been realized, e.g., complex micro lenses (8, 30–32), gradient index lenses (33), diffractive lenses (34), and other wavefront modulation elements working in the visible band (35, 36). The crosslinked resin in TPL is polymeric acrylates with a low refractive index (LRI) of ~ 1.5 , negligible absorption, and an Abbe number of ~ 35.2 (for the IP-Dip resin used here) (37). Hence, the sensitivity to changes in wavelength is smaller compared to the

more commonly used HRI materials in metalens design. Thus, TPL patterning with LRI polymers mitigates the material dispersion problem and enables high-resolution fabrication of complex 3D structures required in MAMs. Its ability to pattern subwavelength features to modulate the wavefront have been demonstrated with applications in the visible and IR range (38–41).

In recent days, metadevices with novel functions or multifunctionality, and the design process are benefiting from the development of inverse design and intelligent design algorithms (42–44). Here, we used topology optimization (TO) with the adjoint method to obtain suitable design parameters to achieve achromatic lensing behavior. This approach enables inverse design of nano-structure heights within each layer at varying distances between layers and number of layers to achieve high-efficiency metalenses. By using the optimized geometric parameters and the inverse designed structure, we tailored our fabrication process to achieve a stable, multilayer, and high-resolution structures in a short fabrication time of ~ 1 min for each MAM. Using shrink-compensation

and proximity-effect reduction techniques, we successfully 3D-printed the MAM (schematic in Fig. 1A) with 400-nm bandwidth ranging from 400 to 800 nm, NA of 0.5 and 0.7. We discussed the effect of edge rounding and smoothing due to limited resolution during 3D printing and their influence on the MAM performance. Simulation results agree with the measured achromatic performance. Last, we demonstrated the performance of the MAM by imaging a resolution target with white light illumination. The 0.5- and 0.7-NA MAMs achieve an average focusing efficiency of 42% and 20%, respectively. Here, the focusing efficiency is defined as the ratio of the power at the focal spot within the circle with a diameter of $3 \times \text{FWHM}$ (full width at half maximum) to the total incident power on the MAM (45).

The MAMs demonstrate previously unachievable performances in efficiency, bandwidth, and NA as shown in Fig. 1D, in which the figure of merit is defined as the square root of the sum of the squares of three parameters. Compared with reported achromatic metalenses, the MAMs overcame limits in bandwidth and NA, as shown in Fig. 1D. Our work integrates the advantages of LRI materials, nanoscale high-resolution 3D printing, and TO to achieve MAMs with exceptional performance, which may inspire a new paradigm for the design and fabrication of multifunctional broadband optical elements and devices.

RESULTS

MAM design

The principal difference between a multilevel metalens and a multilevel diffractive lens (46) is the size of the smallest feature. Here, the minimum feature size is designed to be 0.2 μm ; the 0.2- μm width is set by the resolution limit of TPL achievable in our laboratory with our equipment. Thus, it requires full-wave simulations to account for interlayer interaction and scattering. The proposed model is shown in Fig. 2A for the diametric cross-section of the circular MAM. The darker regions in Fig. 2A are the solid parts of the MAM consisting of crosslinked IP-Dip resin. In the model, $R = 10 \mu\text{m}$ is the lens radius with a 2- μm -thick ring used as structural support. The designed focal length $f = \frac{R}{\tan(\sin^{-1}\text{NA})}$. The thickness of the supporting layer $h_s = 0.5 \mu\text{m}$, and h_i is the distance from the bottom of the n th supporting layer to the top of $(n - 1)$ th layer beneath it. To capture most of the light from one layer to the next, i.e., minimizing losses caused by radial diffraction, h_i is set to be less than 2 μm (fig. S2). Ω_n is the design region consisting of 20- μm -wide and 1.2- μm -tall n rectangular regions, one for each corresponding layer. In Ω_n , we defined 100 rectangles with 0.2 μm width, each corresponding to the cross section of a ring around the lens. Due to axial symmetry, we only need to optimize for the heights of 50 rectangles. The maximum height of each ring is chosen to be 1.2 μm according to simulations showing that rings of heights 0 to 1.2 μm high will achieve the requisite geometric phase of 0 to 2π . The height of the rectangle was discretized to 31 levels (30 blocks along the z axis). In design region Ω , a tensor ξ (n , 50, 31-1) can be used to represent all the variables. Their values range from 0 to 1, where "0" corresponds to air and "1" corresponds to resin. Other values will use the effective refractive index to represent. In addition, we take filtering and binarization steps before we turn the designed structure into a real structure. We varied h_i from 0 to 2 μm and number of layers from 2 to n , respectively, in each

optimization round. In the p th round, we optimized and obtained the variable tensor ξ_p according to h_{ip} and n_p .

For a single wavelength, the optimization goal in designing a focusing metalens is to maximize the intensity $\Phi = |E(\xi)|^2$ at the focal spot of the lens at the specific wavelength. For a broadband case, where the goal is to increase the overall focal intensity, the strategy is to maximize the minimum focal intensity across all wavelengths. Thus, we have the following function (47)

$$\max\{\min[\Phi(\xi, \omega)]\} = \max[\min(|E(\xi, \omega)|^2)] \quad (2)$$

Here, the TO with the assist of adjoint method is applied on the material distribution and geometric configuration to achieve the convergence of the functions above. TO algorithm optimizes the material layout within the user-defined space for a given set of conditions and constraints. To solve the multilayer TO problem, the parameters for all layers should be included in one integrated tensor ξ to be optimized simultaneously (44, 48, 49). Thus, the broadband metalens optimization goal can be written as

$$\max_{\xi(r)}\{\min[\Phi(\xi, \omega)]\} \quad (3)$$

$$s.t. \varepsilon_r(\xi) = \varepsilon_{r, \text{IP-Dip}} + \xi \cdot (\varepsilon_{r, \text{IP-Dip}} - \varepsilon_{r, \text{Air}}) \quad (4)$$

$$0 \leq \xi \leq 1 \quad (5)$$

Here, $\varepsilon_r(\xi)$ is the position-dependent effective permittivity profile used in the inverse design process. The full inverse design process is illustrated in fig. S3. As mentioned before, in each iteration, we optimized and obtain the variable tensor ξ for different h_i and n . With additional layers beyond a certain point, we expect the overall efficiency to decrease due to multiple interfacial reflections and scattering. In addition, with smaller spacing distance (sp), the stronger interaction between layers tends to reduce efficiency, while larger distances result in high-order diffraction and more energy losses. Note that the optimized performance for the single-layer metalens was poor, as expected from the single-layer limitation between NA and bandwidth (results in fig. S4). Thus, as shown in Fig. 2B, we searched through the parametric space with layer number n from 2 to 6 and spacing distance h_i from 0 to 2 μm (0, 0.25, 0.5, 1.0, 1.5, 2.0, 2.5) initially for MAM with 0.5 NA. The efficiency starts to decrease after three layers and drops dramatically when the number of layers reaches six, due to the diffraction away from the optical axis between layers.

We found an initial optimal performance at three layers and a spacing distance in the range from 1 to 2 μm . Setting $n = 3$, we ran another round of optimization for spacing distance h_i in the range of 1 to 2 μm with a step of 0.1 μm . The final design has three layers and a spacing distance of 1.6 μm , which shows a peak in normalized intensity from the approximate fitted parametric exploration curves in Fig. 2B. Although the above process was optimized for MAM with 0.5 NA, it is also applicable for the 0.7 NA case to achieve an optimal achromatic design (details in fig. S5).

Influence of print resolution

The samples were fabricated using the Nanoscribe GmbH Photonic Professional GT2, a TPL-based 3D printing system using a galvo-scanned focused beam from a 780-nm femtosecond laser to

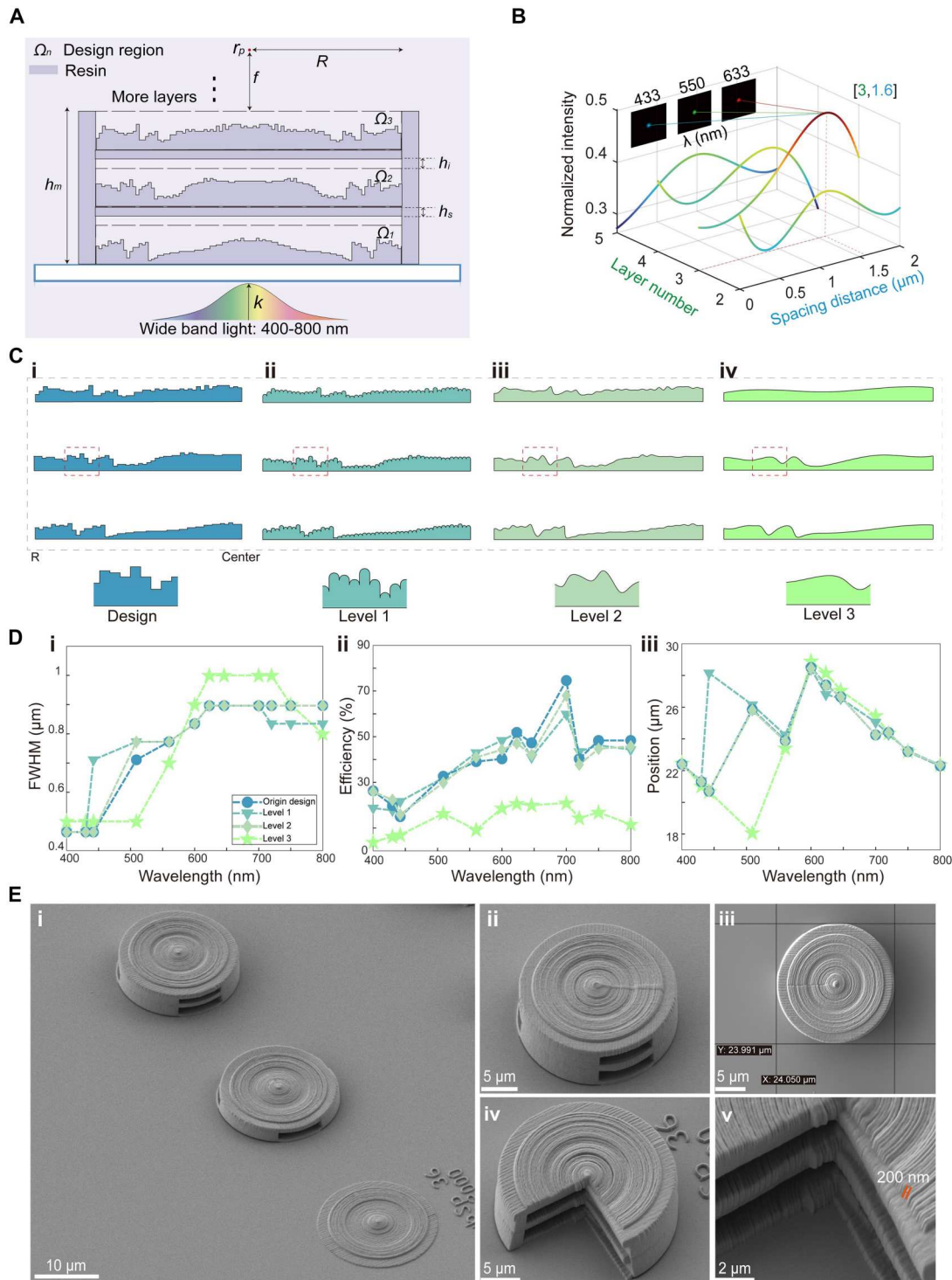


Fig. 2. Topology optimization of the MAM with different layer number and spacing distance. (A) Design model and schematic of the optimization region with indicated parameters described in the text. (B) Relations of the normalized intensity with the layer number and spacing distance. With the inverse design, the best case is located at $[l, sp] = [3, 1.6]$. (C) Schematic of the edge rounding and surface smoothness approximations at different levels, initial design (i), level 1 by rounding the top (ii). Level 2 is generated by applying 10-nm relative tolerance interpolation to the origin height vector (iii), and level 3 is generated by applying 25-nm relative interpolation (iv). (D) Calculated FWHM (i), efficiency (ii), and position of maximum focal intensity along the propagation axis (iii) for different levels. The efficiency (ii) is calculated at the focal plane corresponding to the maximum focal intensity. (E) Tilted view SEM images of the fabricated MAM with 0.5 NA: (i) deconstructed MAM showing single, double, and triple (full) layers; (ii) enlarged view of full MAM; (iii) top view and size of the MAM; and (iv and v) sectioned MAM revealing internal structure and details of the 200-nm-wide ring structures.

induce crosslinking of a liquid resin into a nanoscale solid voxel at the focal spot (details in Materials and Methods). One voxel is an ellipsoid that takes the shape of squared intensity profile of the focused Gaussian beam of the laser (50). As we printed at the resolution limit of the TPL, the fabricated structures will inevitably exhibit deviations from the nominal design of Fig. 2C, i. Studying these deviations, such as corner rounding and loss of details, can inform us about the fabrication process and improve our design approach. In Fig. 2C, we introduced corner rounding by introducing a fillet to the top edges while preserving the structure heights, resulting in Fig. 2C, ii. To introduce a more severe effect of variation on the surface profile, we adopted the linear interpolation to Fig. 2C (iii and iv) with different relative tolerance, and the relative tolerance means the maximum allowed changes between the structure height in the newly generated height curve and the corresponding structure height in the sequence of heights we input. We interpolated the original design (height curves) with a relative tolerance of 10 nm, generating the structure depicted in Fig. 2C, iii. The topography is smoother while retaining most details. Similarly, using interpolation with a tolerance of 25 nm, we obtained Fig. 2C, iv. Within Fig. 2C, we observed the disappearance of sharp edges from Fig. 2C (i and ii). From Fig. 2C (iii and iv), only the overall surface morphology is preserved, with significant loss of details. We conducted simulations of the corresponding MAMs for the four different structures and compared their performance. As shown in Fig. 2D (i to iii), the efficiency and FWHM are almost unchanged for levels 1 and 2 compared to the original design. However, for level 3, where most of the height information is lost, we observed a drastic drop in the efficiency. The fine structures on the surface topography obtained via optimization appear to guide and scatter multiwavelength light in a purposeful manner to achieve achromaticity and are vital for realizing high efficiency for different wavelengths. At level 3 approximation, only the low-frequency (global) phase variation along the radial direction is maintained; thus, the focusing function still works, while the efficiency is the lowest. We extracted the focal point position variation on the propagation axis as shown in Fig. 2D, iii. Except for two data points at 442 nm of level 1 and at 509 nm of level 3, the four different shapes have good consistency in their ability to achieve broadband focusing of light. The reason for the two outliers is the presence of two focal points on the axis of light propagation and the selection of the focal position based on higher intensity (in fig. S6). In Fig. 2D, iii, longer wavelength shows good consistency due to the less sensitivity to small features. If we consider the balance of manufacturing feasibility and performance [we also compared the point spread function and modulation transfer function (MTF) of the original structure and level 1 structure, which is included in fig. S7; original structure and level 1 structure show good consistency on the MTF performances], original design and level 1 are the better choices. In the next section, based on this result, we optimized the fabrication method to achieve a prototype close to the nominal design.

Experimental results

With the optimized parameters, we experimentally demonstrated MAMs with NA at 0.5 and 0.7, having high focusing efficiency with broadband achromatic performance. By using TPL, we precisely realized the designed MAM with nanoscale feature sizes. As the width of the ring measured only 200 nm, overcoming proximity effects during printing of the dense rings becomes a challenge,

especially in the central region, which would be overexposed and end up being taller than designed. To circumvent this effect, we introduced a 500-ms interval between the exposure of adjacent rings and exposed the rings in sequence starting from the center and moving toward the edge. A detailed study of this procedure is beyond the scope of this work. In addition, we optimized the interval of printing in the z direction for different height structure, further improving the precision. For each layer, the outer circular support was first printed with rings, and then the support layer was printed with straight bridging lines to increase mechanical stability. To maintain the specific distances between different layers, the surrounding wall is necessary to support them firmly. Small openings were added on the sidewalls to allow drainage of uncured liquid resin from between the layers of the MAM during the postdevelopment process. These openings were distributed evenly with a 120-degree angle from one layer to another to maintain structural integrity, which can be seen in Fig. 2E (ii). Because of the limited degree of crosslinking of resin during exposure, 3D-printed structures usually suffer from shrinkage. Thus, in addition to using postdevelopment ultraviolet (UV) exposure to increase the degree of curing, we also compensated for shrinkage by increasing the nominal size of each layer to achieve the intended size of the design. We first printed the structures without compensation to determine the volumetric shrinkage from scanning electron microscopy (SEM) inspection. Afterward, successful MAMs were obtained in the final fabrication (Fig. 2E, i to v), with compensation added by scaling the three layers by a factor of 1.01, 1.02, and 1.03 from the bottom to the top (details in fig. S8). We also printed a separate structure with a section removed (Fig. 2D, iv and v) to verify the dimensional agreement with the design and observe the good separation between individual rings and layers within the MAM.

We characterized the achromatic focusing performance of the MAMs by measuring the distribution of light intensity in the focal plane (xy) and along the propagation plane (xz) with a super-continuum laser source. As seen in Fig. 3 (A and B), the lengths of the focal spots at different wavelengths are less than 2 μm , and the focal positions shift by less than 3 μm across the visible spectrum. We observed good agreement between simulated and measured results for the intensity distributions at the focal and axial planes for both 0.5- and 0.7-NA MAMs at six discrete wavelengths of 442, 509, 560, 600, 623, and 646 nm, as shown in Fig. 3. Measurements beyond the range of 442 to 646 nm were not conducted due to limitations of our complementary metal-oxide semiconductor (CMOS) sensor. The experimental focusing efficiency across the wavelengths ranges from 22.5 to 42.7% for 0.5 NA, with corresponding simulated efficiencies ranging from 24.7 to 74.2%. For the 0.7 NA, the experimental focusing efficiency across the wavelengths is in the range of 13.3 to 22.1%, appearing to closely match the corresponding simulated efficiencies of 14.6 to 26.1% (Fig. 4, A and B). The inconsistency between measurements and simulation is likely due to imperfections in the fabricated lenses as explained previously, a non-ideal collimated light source, and stray light. The wavelength averaged simulated focusing efficiency is 44.2 and 20.0% for 0.5 and 0.7 NA, respectively. The simulated FWHM is ~ 0.88 to 1.4 μm for 0.5-NA MAM and 0.64 to 1.8 μm for 0.7-NA MAM.

We assessed the imaging quality of the MAM with NA of 0.5 by placing it onto a USAF 1951 resolution target with $\sim 100\text{-}\mu\text{m}$ spacing between the top of the MAM and the top of the resolution target. A

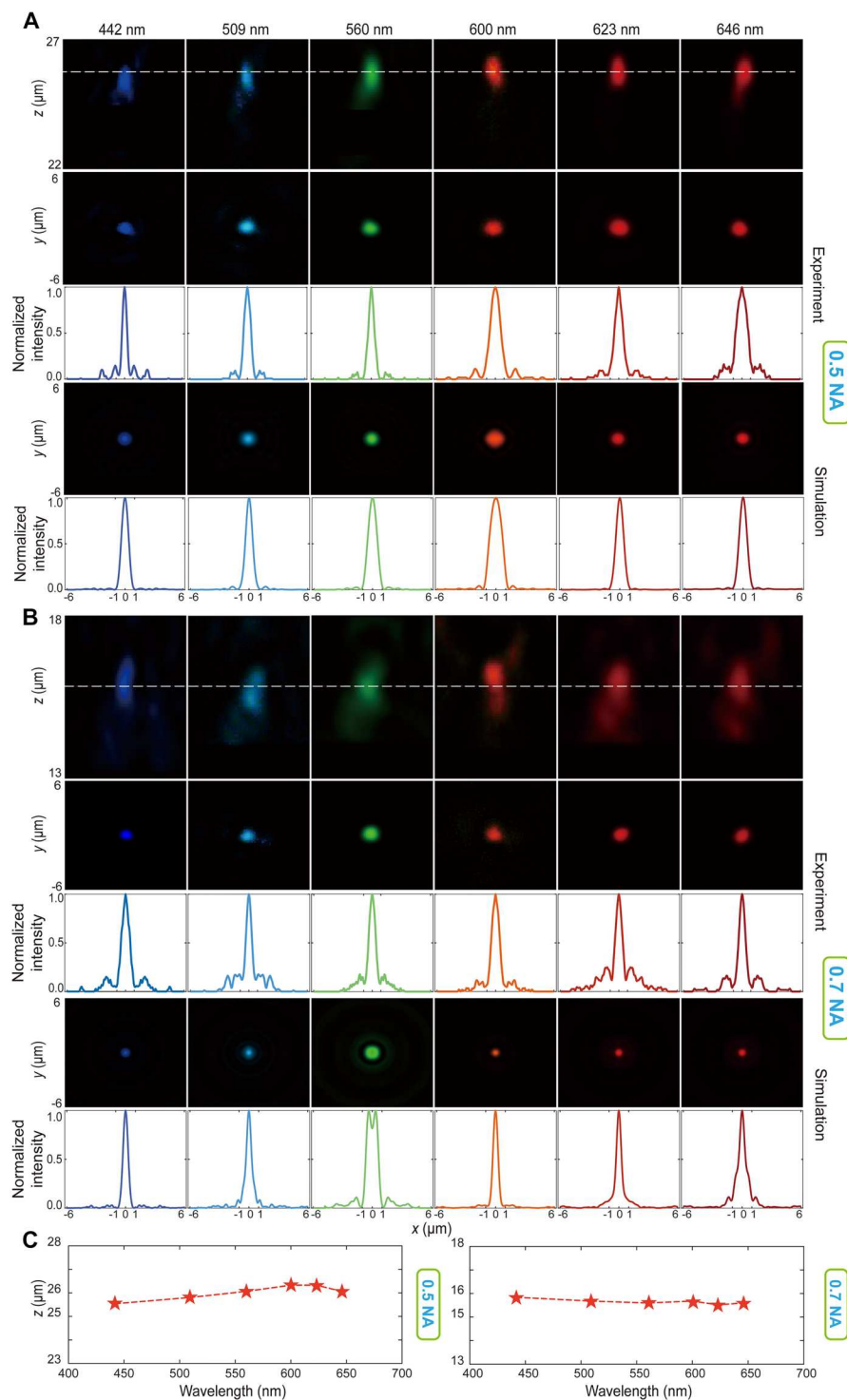


Fig. 3. Characterization of achromatic performance of the fabricated MAMs. Intensity distributions on axial and focal planes for MAMs with (A) 0.5 NA and (B) 0.7 NA. In both panels, the first row is the experimentally measured normalized focal length shifts in the xz plane, z axis distance measured from the top of the sample, the second row is the measured intensity distributions in the xy plane, and the third row is the normalized intensity along the white dashed lines at the focal plane. The fourth and fifth rows are the corresponding simulation results (details in fig. S5). (C) Measured focal spot positions along the z axis for MAMs with 0.5 and 0.7 NA.

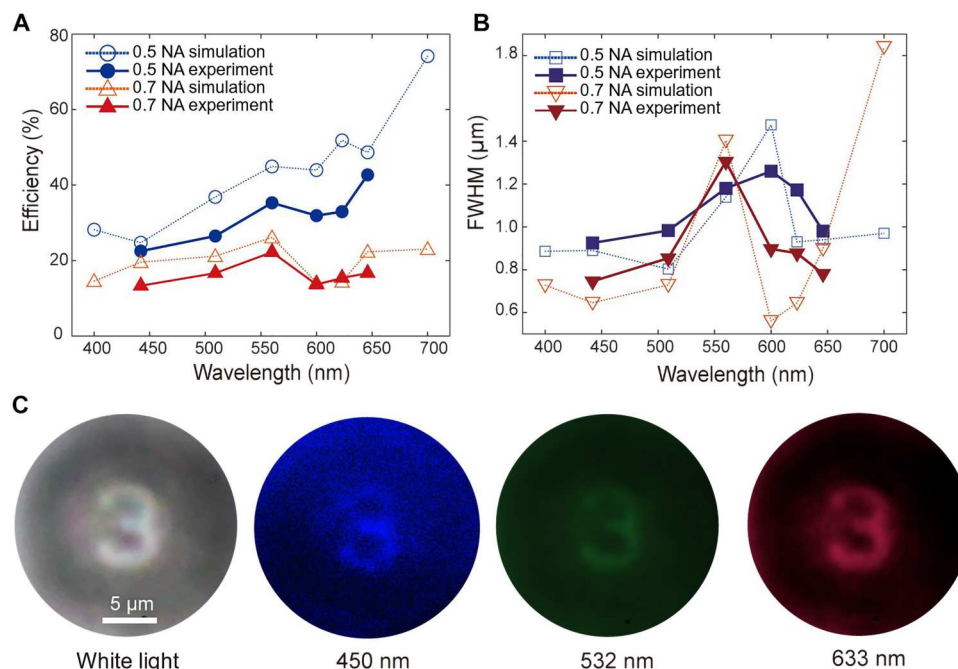


Fig. 4. Focusing efficiency and imaging performance of MAM. (A) Comparison of experiment and simulated broadband focusing efficiencies for MAMs with NA of 0.5 and 0.7 at the same focal plane defined by NA. (B) Comparison of experiment and simulated broadband FWHM for MAMs with NA of 0.5 and 0.7 at the same focal plane defined by NA. (C) Optical images of the number “3” in group 6 element 3 in the USAF 1951 resolution target captured through the 0.5-NA MAM under white light and applied blue (450 nm), green (532 nm), and red (633 nm) filters.

minimized image of group 6 element 3 in the resolution target was imaged through the MAM, which has a spacing distance of three times the focal length to the objectives (18). The images were captured under a microscope under white light illumination and with color filters at wavelengths of 450, 532, and 633 nm placed in front of the white light source as shown in Fig. 4C. The results demonstrated that the fabricated MAMs performs well under incoherent white light for achromatic imaging applications (more images can be found in fig. S9). We also fabricated a 0.5-NA spherical lens and a Fresnel lens for comparison to demonstrate the unrivaled ability of the MAMs in removing chromatic aberration (see fig. S10).

DISCUSSION

In a multilayer metalens system, each layer can be regarded as an achromatic corrector and focusing element. The MAM overcomes constraints that trade-off between NA and bandwidth in single-layer achromatic metalens. While three layers enables high efficiency, MAMs with higher NA can be achieved by adding layers to the system at the expense of efficiency. The effect of spacing distance on the interaction between layers is difficult to quantify with a physical model, and thus, efficient and accurate inverse design methods, like the TO presented here, are particularly useful to design high-performance structures. Currently, the optimization of multilayer structures requires full-wave simulations, making it challenging to implement mathematical analysis or approximations in electromagnetic calculations, e.g., Rigorous coupled-wave analysis (RCWA) or locally periodic approximation to accelerate the inverse design process. Algorithms using approximations and avoiding solving Maxwell equations for each step should be used for designing

multilayer meta-devices of larger dimensions. We found that for distances smaller than 0.5 μm, interlayer coupling greatly reduces the transmitted power. Conversely, if the spacing distance is larger than 2 μm, high-order diffraction losses also decrease the efficiency. Hence, we performed a high-resolution search within these limits to find an optimal spacing distance of 1.6 μm. Our fabricated MAMs with broadband operation in the wavelength regime of 400 to 800 nm and average focusing efficiencies of ~44.2% for 0.5 NA and 20% for 0.7 NA demonstrate the potential of the inverse design and TPL fabrication methodology. This work pushes the boundary of metalens design to the elusive region where high NA, broadband operation, and high achromatic efficiency can be achieved simultaneously.

Our results demonstrated that stacked metasurfaces based on LRI materials overcome the limitations of single-layer flat optics to extend the performance of metalenses into broadband operation while preserving high NA. We harnessed the power of TO and adjoint method to accelerate the iterative design of metalenses, and used it to overcome the design constraint of good achromatic performance with simultaneously high NA. The presented design framework and experimental fabrication by TPL at a resolution of ~200 nm provide a practicable path from 2D metasurfaces to complex 3D metasystems. The ability of TPL to fabricate structures on various substrates such as fiber facets, CMOS, or surfaces with curvatures extends the application scenarios of our MAM. In the future, with higher-resolution 3D printing techniques and HRI resins, we envisage an increase in multifunctional optical systems that operate with a broadband response beyond the visible range.

MATERIALS AND METHODS

Finite-difference time domain simulations

The full-wave simulations of the MAMs were carried out with numerical finite-difference time domain, and the adjoint method algorithm was conducted by the numerical script. In the adjoint method process, a 2D cross section of the MAM in the xz plane was chosen for the simulation model as shown in Fig. 2B. The boundary conditions are set as Perfectly Matched Layer (PML). The light source constituted of both x and y polarized light sources with the same amplitude to represent unpolarized light. The TO algorithm was conducted with an in-house MATLAB code.

Fabrication with TPL

The fabrication of MAM was performed using a TPL-based 3D printing technique (Nanoscribe GmbH Photonic Professional GT2). A 63 $\times$ objective lens with NA = 1.4 was used in dip-in laser lithography mode. For the fabrication of the supporting layer and supporting pillars, ContinuousMode in GalvoScanMode was applied with the PiezoSettlingTime of 50 ms and GalvoSettlingTime of 4 ms. For the fabrication of the metalens structures, the PiezoSettlingTime was changed to 500 ms and GalvoSettlingTime remained as 4 ms. The printing parameters for the supporting layer and supporting walls were ScanSpeed of 6000 $\mu\text{m/s}$ and LaserPower of 40% of the total power (50 mW). For the center part of the MAM structures (from the center to the 10th rings), ScanSpeed was 60,000 $\mu\text{m/s}$ and LaserPower was 30%, corresponding to 15 mW. For the left MAM structures (from the 11th to the 50th rings), ScanSpeed was 40,000 $\mu\text{m/s}$ and LaserPower was 32%, corresponding to 16 mW. The waiting time between printing two rings was set as 500 ms. After the exposure process, the sample was developed in propylene glycol monomethyl ether acetate for 6 min, isopropyl alcohol for 5 min, and methoxy nonafluorobutane for 7 min. When emerged in isopropyl alcohol, the sample was cured with UV light for 5 min with 70% maximum power (DYMAX, MX-150, 405 nm).

Characterization

A supercontinuum light source (SuperK SELECT UV/nIR1, NKT Photonics) was used for the characterization of achromatic performance within the continuous bandwidth of 450 to 650 nm. The optical testing apparatus is included in fig. S11. The laser passed through a pair of periscopes and collimated along the optical path with lenses. The focal plane of the MAM was imaged by a 150 $\times$ objective lens, and then the image was captured with a high-resolution CMOS camera (DCC3260C, Thorlabs). The objective lens was mounted on a piezo stage (P-736.4CA, PI) with increment of 20 nm, and then the distribution of focused light near the focal spot was taken by moving the piezo stage with step of 20 nm. To measure the focal efficiency, an additional aperture was placed in front of the camera with the aperture tuned to the size that we can only observe the focal spot in the camera. Afterward, the camera was substituted by a photodiode power sensor (S142C, 350 to 1100 nm, Thorlabs) to measure the power at different wavelengths. The total incident light power on the MAM was measured with the same configuration, but with the aperture set to the size of the MAM image after the imaging system, and with the MAM removed.

Imaging

The USAF 1951 resolution target was fabricated with EBL (Raith, eLINE Plus)-based patterning and Cr deposition with electron beam evaporation (Kurt J. Lesker), followed by a lift-off process. The MAM sample was placed onto the resolution target with the MAM facing down. A tape with thickness of around 100 μm was used at the edges of the glass substrate to fix the relative position between MAM and target. The image was taken with a microscope (Nikon Eclipse LV100ND) in transmission mode with a 50 $\times$ NA 0.4 objective lens. By reducing the aperture size of the halogen lamp, the directivity of broadband white light source was better for imaging test. The clear imaging was obtained by tuning the focal plane of microscope away from the top surface of the target plane to the designed image plane. To evaluate the imaging quality at the wavelengths of 458, 532, and 633 nm, corresponding color filters (FL457.9-10, FL532-10, and FL632.8-10, THORLABS) were placed on the top of the halogen lamp.

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S11
Tables S1 and S2
References

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Acknowledgments

Funding: This work acknowledges funding support from the National Research Foundation (NRF) Singapore, under its Competitive Research Programme award NRF-CRP20-2017-0004 (J.K.W.Y.) and NRF Investigatorship Award NRF-NRFI06-2020-0005 (J.K.W.Y.). **Author contributions:** Conceptualization: C.-F.P., Hao Wang, and J.K.W.Y. Methodology: C.-F.P., Hao Wang, C.-W.Q., and J.K.W.Y. Fabrication: C.-F.P., Q.R., J.Y.E.C., W.Z., and Hao Wang. Characterization: C.-F.P., Y.K., and Hongtao Wang. Simulation: C.-F.P. Schematic drawing: C.-F.P. and P.N.S. Supervision: C.-W.Q. and J.K.W.Y. Writing (original draft): C.-F.P. and Hao Wang. Writing (review and editing): C.-F.P., Hao Wang, S.W., C.-W.Q., and J.K.W.Y. All authors discussed the results and commented on the manuscript. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials.

Submitted 25 July 2023

Accepted 21 November 2023

Published 20 December 2023

10.1126/sciadv.adj9262